The HPA axis, better known as hypothalamic-pituitary-adrenal axis, plays a central role in regulating stress. The HPA is part of the neuroendocrine system that consists of the hypothalamus, pituitary gland, and adrenal glands. When the body is under stress, the HPA axis is activated. The hypothalamus is activated first by sending a signal called the corticotropin-releasing hormone to the pituitary gland; next the pituitary gland releases adrenocorticotropic hormone to the adrenal which releases cortisol into the body. When receptors in the brain such as the hypothalamus and hippocampus senses high cortisol levels in blood, the HPA axis will become deactivated (negative feedback). The release of cortisol helps increase the supply of glucose to help the body cope with the anticipated stress. The HPA axis becomes dysregulated when activation or deactivation of the HPA axis becomes out of order. A dysfunctional HPA axis produces abnormal levels of cortisol.

HPA serves an important role in our physical health and mental health. Preparing the traumatic stress response of fight, flight, freeze to better prepare the body to take action when faced with stressful situations such as danger. The cortisol released by the HPA axis plays a role in the survival and death of neurons, changes to anatomical brain structures, forming new memories, and emotional interpretation of events (Herbert et al., 2006).

There is a strong association between HPA, anxiety and post-traumatic stress disorder. In addition to the link between dysregulated HPA axis and psychiatric disorders, dysregulated activity of HPA can be linked to diabetes, hypertension, cancer, and cardiovascular disease (McEwen,1998). Repeated stress has consequences on brain functioning, particularly the hippocampus, which affects memory. The hippocampus can play a role in perceiving events as threatening or nonthreatening (McEwen, 2006). Dysregulation of the HPA axis and depression has been linked to Cushing's syndrome which is associated with prolonged high cortisol levels in the body.

Researchers have found that the HPA axis is abnormal in psychiatric disorders, particularly in major depression (Pariante & Lightman, 2008). Depressed patients have increased cortisol in the saliva, plasma, and urine, and increased size and activity in the pituitary and adrenal glands (Korosi & Baram, 2008). Increased levels of stress are linked to changes in the HPA axis and dysregulation of the serotonin pathway as well as the gene linked to both suicide and mood disorder. Regardless of psychiatric condition, HPA dysregulation has been linked to suicide (Berardelli et. al, 2020). If the HPA axis is continuously dysregulated, it may result in damages in the HPA axis as well as bodily tissues leading to various physical and mental health ailments (McEwen, 1998).

The relationship between HPA and suicide is evidenced by the failure to suppress cortisol among patients with mood disorders who died by suicide (Coryell and Schlesser, 1981), but the timing of stress levels on suicide is still unknown. To test the information regarding how the HPA axis responds to stress, the majority of studies use pharmaceutical drug induction to mimic stress. One pharmaceutical test called the Dexamethasone Suppression Test, DST, suggests that HPA hyperactivity is consistent with completed suicide, rather than suicide attempts (O'Connor, D, 2015). Even though the DST has contributed immensely to research, findings on HPA dysregulation and suicidality using DST have been inconsistent and contradictory (McGirr et al., 2011). Some have found the HPA hyper-activity in in-patients but not out-patients (Coryell et al., 2006) while others in male but not female (Jokinen et al., 2009). Some studies reported there is association between HPA dysregulation and suicide (Jokinen et al., 2010) while others didn't find any link between HPA dysregulation and suicide (Dahl et al., 1992). Cortisol has been found to be linked with impairments in cognitive functioning such as decision-making and emotional processing all of which are factors of suicidal behavior (Turecki et al., 2012). In addition to aspects of emotion dysregulation, there is evidence that supports the physiological component of suicidality, such as noradrenergic overactivity from HPA dysfunction (Heim & Nemeroff, 2001). These studies demonstrate that the connection between depression and suicide are present with respect to both brain functioning and abnormalities in brain regions that serves those functions.

In terms of treatment, changes to a healthier lifestyle and habit can help reduce stress. Knowing how to regulate emotions may lower stress levels. Being able to get high quality sleep, exercise, as well as eating a well balanced diet. Sleeping pills, anxiolytics, beta blockers, antidepressants may also be helpful (McEwen & Gianaros, 2011). Surprisingly, with the usage of antidepressants, the impairment of the hyperactive HPA axis can be resolved (Gardner et al., 2005). Just as taking antidepressants can temporarily improve the effects of depression related to abnormalities in brain anatomy, serotonin levels, and mood, the same can be applied to dysregulated HPA. While common sense of changes to a healthier lifestyle can help manage stress, it is also important to know that how one perceives an event can also have an effect on stress and mood. Take for example, if a student perceives that they didn’t choose the correct major they may stress on the issue of being successful in terms of their career, whereas if they look outside the box that failure is something that leads to success then they will be less likely to be overwhelmed and stressed.

In conclusion, the evidence is strong for the relationship between HPA dysfunction and depression. Physical evidence such as abnormalities in the serotonin system, norepinephrine, and brain anatomical structure shows the genetic vulnerability intersecting with environmental factors. The hyper-activity of pituitary and adrenal glands, increased cortisol in saliva, plasma and urine measured in depressed patients furthers this evidence of a connection between depression and HPA dysfunction. One of the HPA components is the hippocampus, which is responsible for memory formation and interpretation of events, and is often affected by the HPA dysfunction. Memory and appraisal of events are impaired when someone is experiencing depression, furthering the link between HPA dysfunction and depression.

Despite the strength of this evidence, there are some inconsistencies associated with age, time of data collection, psychiatric conditions, methodology, as well as genetics. Pharmacological manipulation has been criticized for not being able to mimic the real extent of real- life stressors (Burke et al., 2005). In recent years, there has been an increase on research between suicide and the HPA axis, though much remains unknown. The relationship between hyper- and hypo- activity on the HPA axis in terms of mood disorders is not very clear, particularly the timeline between hyper- or hypo- activity and a depressive episode. Likewise, the levels of cortisol before and during a suicidal episode is unknown. The specific age where cortisol levels diminish were also not known as one study only chose 40 as the age for comparison. Many findings are inconclusive and contradictory. Future directions should include finding a test to mimic cortisol more effectively. Rather than just having DST as the single methodology to test the HPA response to cortisol, having more than one single test should be considered. Future studies may also wish to more closely assess the timeline for cortisol changes and episodes of suicidal ideation.

My hypothesis for this paper was that the dysregulation of the HPA axis is associated with depression. The inspiration behind this paper was while studying the HPA axis in Abnormal Psychology, I was curious how might the HPA be involved with various psychiatric conditions. I would imagine a person who is suicidal would be under extreme stress levels and pain. Knowing this relationship between depression and suicide, I wanted to know more how the HPA ties into them both. The majority of the content in the field of psychology focuses mainly on the emotional aspects. I wanted to not only understand the relationship but also the physiological component as I feel understanding psychology both in terms of emotion and physiology gives a more significant impact to deepen our understanding.

While researching, everything started making sense. Pieces of my prior knowledge started to come together. Even though I didn't include the serotonin process into the analysis part of this paper, I learned that serotonin is mainly produced in the gastrointestinal system whereas 10% of serotonin is made in the brain. This is the reason why being malnourished can produce some devastating effects on your mentality. The lack of nutrition can disrupt the serotonin system, this implies the link between eating disorders and various mental disorders such as depression. Another point where I thought was very interesting was the fact that medications such as ones for depression can reverse the effects of the dysregulation of HPA. While researching information on HPA, I came across a research paper on the sex differences in the brain. It is well known that the risk for females is much higher than for males in the majority of psychiatric disorders. During lectures, the explanation that females face a much higher stress rate with contributing factors such as increased rates gender discrimination, rape, giving birth and so on. I never quite understand why females face a higher risk for psychiatric disorders even with that given explanation. If that is the case, considering globally and the diversity of cultures it cannot be simply because of the stress on females as each region and culture has different sets of expectations. Though I haven't dug deeper into that article, there have been well known sex differences in the brain in terms of anatomy. I think this explanation is a bit more satisfying for my understanding. There has been a saying that is a theme in the STEM field, the female and male brains are wired differently so females are highly desired in the tech field because it is a dominantly male field there is a lack of diversity especially in terms of thought process. I never really cared much as I just take words with a grain of salt, but upon coming across this paper, I understood there was actual evidence behind it.

After reading the article on evolutionary psychology, I learned the majority of our genes change depending on our environmental factors. The changing of gene expression helps us better evolve and survive. This is also another reason why the trait diathesis exists. I also thought more about the purpose of suicide. I remembered watching a documentary on honey bees. The colony of bees is becoming endangered, particularly honey bees because of being infected with viruses or parasites. When a bee is sick, the protocol is to leave the nest as they don't want the other bees to be sick thus dying alone. This type of behavior in a sense is similar to suicide that is the wanting to die alone. As I ruminate more on this topic I thought more between the article and the honey bee connotation. It's interesting how Adam had brought up a point where survival of the fittest can also be in terms of a whole rather than individualistic. I had never thought about thinking in terms of a pack. Maybe it is the individualistic culture rather than the collective society that influences such thinking. Even though we live in an individualistic society, I think we should take into consideration all perspectives to get a bigger picture understanding. I thought this article was very interesting as I learned a lot of new information.

From my perspective, the task that I highly valued as well as hated was the transcription process. The transcription process was relatively labor intensive. The interviewees were very insightful and vulnerable. Interviewees mainly expressed their stress on academics. I didn't really notice such a trend until Adam pointed it out to me. The content seemed very much normalized but I realized it may just be me. I'm pretty desensitized to having life being outside of academics as it was how I was raised. It is also to note that some people may not be as comfortable with sharing more vulnerable personal information with a stranger who you just met. Being a first-year student, especially if coming from out of state or just not having your childhood friends go to the same university as you, can be a factor of isolation. One of the harder parts of transcribing was that I was listening to sections and then stopping to clean the transcriptions so I was not able to really listen without interruptions. To be clear, the stopping then listening and stopping again disrupts some of that understanding of the conversation contents. Another thought that came to mind while transcribing was, if people could be safely vulnerable where others can create this safe space, wouldn't it be nice how everybody can relate as we all have some level of similarity in terms of experiences. Coming from the side of simply being a listener or observer, I finally understand the pain of not being able to have a voice of saying I relate. Just because the listener is not saying anything doesn't mean that they don't have any reactions. It's always been so easy to assume that everybody has everything together is normal and healthy when in reality, all is just a delusion.

At the big lab meeting in July, I thought it was very fascinating to see how the other members' perspective comes into play. Although I didn't speak out on my perspective, there was a reason behind it besides the fact of being relatively new and facing imposterism. Sometimes speaking rather than listening, you won't be gaining as much of a diverse perspective. Usually whenever a person starts a trend, it will continue with that trend because people get fixated on someone else's opinion forgetting that there’s other perspectives to take into account as the topic is framed around that trend. I think this was a special case of this happening. After that meeting, I thought about what if I did share my opinion? Will my opinion spark others to follow a trail or will this create a new divergence of thoughts that differ from mine? That is, the reason I did not speak was the fear of people following a trail instead of diverging into a diversity of thoughts and opinions. The big takeaway from this snippet is to be divergent rather than staying silent because nobody knows what will happen when something new comes into play.

Continuing with the July’s meeting, other researchers in YDSP agree with the fact that mobile intervention apps are often a failure. Even looking at the data collected, it is agreed this project is going to fail as users don’t view mental health apps as positive. I agree, but at the same time I disagree. Instead of looking solely on numbers and one-sided story, I see it in another perspective. I’m grateful that I got a chance to take part in the laborious transcribing process, it was where I got to not only get a chance to interact with the content of this project but also expand on my point of view. During the interviews, interviewees did mention why they don’t like mental health apps. The reason why they don’t like these types of apps is that the majority of mental health apps are poorly designed for the users. It is not that they are not interested in the idea of mental health apps or using it, but it’s that the mental health app market is not well catered to users. In fact, interviewees had expressed their voice of wanting a well-designed app for mental health. Just because the data shows one side of the picture doesn’t mean it should be painted in such a way but since we have the interviews, it is another piece of information that should be taken into consideration in addition to the statistics. The moral of the story is that even though data is data, it does not tell the whole story.

References

Berardelli, I., Serafini, G., Cortese, N., Fiaschè, F., O'Connor, R. C., & Pompili, M. (2020). The Involvement of Hypothalamus–Pituitary–Adrenal (HPA) Axis in Suicide Risk. *Brain Sciences*, *10*(9), 653. https://doi.org/10.3390/brainsci10090653

Burke, H. M., Davis, M. C., Otte, C., & Mohr, D. C. (2005). Depression and cortisol responses to psychological stress: A meta-analysis. *Psychoneuroendocrinology*, *30*(9), 846–856. https://doi.org/10.1016/j.psyneuen.2005.02.010

Coryell, W., & Schlesser, M. A. (1981). Suicide and the dexamethasone suppression test in unipolar depression. *American Journal of Psychiatry*, *138*(8), 1120–1121. https://doi.org/10.1176/ajp.138.8.1120

Coryell, W., Young, E., & Carroll, B. (2006). Hyperactivity of the hypothalamic–pituitary–adrenal axis and mortality in major depressive disorder. *Psychiatry Research*, *142*(1), 99–104. https://doi.org/10.1016/j.psychres.2005.08.009

Dahl, R. E., Kaufman, J., Ryan, N. D., Perel, J., Al-Shabbout, M., Birmaher, B., Nelson, B., & Puig-Antich, J. (1992). The Dexamethasone Suppression Test in children and adolescents: A review and a controlled study. *Biological Psychiatry*, *32*(2), 109–126. https://doi.org/10.1016/0006-3223(92)90015-r

Gardner, K., Thrivikraman, K., Lightman, S., Plotsky, P., & Lowry, C. (2005). Early life experience alters behavior during social defeat: Focus on serotonergic systems. *Neuroscience*, *136*(1), 181–191. https://doi.org/10.1016/j.neuroscience.2005.07.042

Heim, C., & Nemeroff, C. B. (2001). The role of childhood trauma in the neurobiology of mood and anxiety disorders: preclinical and clinical studies. *Biological Psychiatry*, *49*(12), 1023–1039. https://doi.org/10.1016/s0006-3223(01)01157-x

Herbert, J., Goodyer, I. M., Grossman, A. B., Hastings, M. H., de Kloet, E. R., Lightman, S. L., Lupien, S. J., Roozendaal, B., & Seckl, J. R. (2006). Do Corticosteroids Damage the Brain? *Journal of Neuroendocrinology*, *18*(6), 393–411. https://doi.org/10.1111/j.1365-2826.2006.01429.x

Jokinen, J., Nordström, A. L., & Nordström, P. (2009). CSF 5-HIAA and DST non-suppression — Orthogonal biologic risk factors for suicide in male mood disorder inpatients. *Psychiatry Research*, *165*(1–2), 96–102. https://doi.org/10.1016/j.psychres.2007.10.007

Jokinen, J., Ouda, J., & Nordström, P. (2010). Noradrenergic function and HPA axis dysregulation in suicidal behaviour. *Psychoneuroendocrinology*, *35*(10), 1536–1542. https://doi.org/10.1016/j.psyneuen.2010.05.008

Korosi, A., & Baram, T. Z. (2008). The central corticotropin releasing factor system during development and adulthood. *European Journal of Pharmacology*, *583*(2–3), 204–214. https://doi.org/10.1016/j.ejphar.2007.11.066

Kudielka, B. M., & Kirschbaum, C. (2005). Sex differences in HPA axis responses to stress: a review. *Biological Psychology*, *69*(1), 113–132. https://doi.org/10.1016/j.biopsycho.2004.11.009

Mann, J. J. (2013). The serotonergic system in mood disorders and suicidal behaviour. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *368*(1615), 20120537. https://doi.org/10.1098/rstb.2012.0537

McEwen, B. S. (1998). Protective and Damaging Effects of Stress Mediators. *New England Journal of Medicine*, *338*(3), 171–179. https://doi.org/10.1056/nejm199801153380307

McEWEN, B. S. (2006). Stress, Adaptation, and Disease: Allostasis and Allostatic Load. *Annals of the New York Academy of Sciences*, *840*(1), 33–44. https://doi.org/10.1111/j.1749-6632.1998.tb09546.x

McEwen, B. S., & Gianaros, P. J. (2011). Stress- and Allostasis-Induced Brain Plasticity. *Annual Review of Medicine*, *62*(1), 431–445. https://doi.org/10.1146/annurev-med-052209-100430

McGirr, A., Diaconu, G., Berlim, M. T., & Turecki, G. (2011). Personal and family history of suicidal behaviour is associated with lower peripheral cortisol in depressed outpatients. *Journal of Affective Disorders*, *131*(1–3), 368–373. https://doi.org/10.1016/j.jad.2010.10.050

O'Connor, D. B., Ferguson, E., Green, J. A., O'Carroll, R. E., & O'Connor, R. C. (2015). Cortisol levels and suicidal behavior: A meta-analysis. *Psychoneuroendocrinology*, *63*, 370–379. https://doi.org/10.1016/j.psyneuen.2015.10.011

Pariante, C. M., & Lightman, S. L. (2008). The HPA axis in major depression: classical theories and new developments. *Trends in Neurosciences*, *31*(9), 464–468. https://doi.org/10.1016/j.tins.2008.06.006

Turecki, G., Ernst, C., Jollant, F., Labonté, B., & Mechawar, N. (2012). The neurodevelopmental origins of suicidal behavior. *Trends in Neurosciences*, *35*(1), 14–23. https://doi.org/10.1016/j.tins.2011.11.008